

Supplementary Material for CauSample

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A Proof of the Causal Sampling Bound

We restate Theorem 4.2 and establish its causal sampling bound through the three steps outlined below.

THEOREM A.1 (CAUSAL SAMPLING BOUND). *For a batch $X_i \in \mathbb{R}^{M \times T}$ and its SCS $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_K\}$, assume each child is an exact linear combination of its parents, whose vectors are linearly independent and span a subspace with incoherence at most μ . Let $\tau = \max_m |Pa(m)| \geq 1$, $l = \max_k |Root_k|$, and $\delta \in (0, 1)$. Then:*

- (1) *Given exact parent data, any child metric can be reconstructed exactly and its causal coefficients recovered with probability at least $1 - \delta$ by uniformly sampling d time steps without replacement, where $d \geq c\mu\tau \log(Kl/\delta)$ for a universal constant c ; if this requirement exceeds T , all entries are sampled.*
- (2) *If $Kl \leq r = \text{rank}(X_i)$, the causal bound is no larger than the global-rank bound under the same incoherence bound: $c\mu\tau \log(Kl/\delta) \leq c\mu r \log(r/\delta)$.*

Proof sketch. The proof establishes how the SCS supports matrix completion through three key steps:

- (1) **Characterize the local rank** (Lemma A.2): show that a child and its parents form a matrix whose rank equals the number of linearly independent parents, and relate this rank to the DAG's roots and the global rank.
- (2) **Prove that sampling preserves this rank** (Proposition A.3): apply a matrix Chernoff bound to obtain a sufficient sample count for retaining the local rank.
- (3) **Establish recovery in the causal graph** (Lemma A.4): show that rank-preserving samples uniquely determine the causal coefficients and recover the child, and distinguish this case from vectors outside the parent span.

We then combine these results to prove the theorem's sampling bound and its comparison with the global-rank bound. After the theorem proof, Corollary A.6 extends the recovery argument to all children by a union bound and induction along the causal orders.

Notation. For a child metric m in batch i , write $P_m = X_i(Pa(m), :) \in \mathbb{R}^{\tau_m \times T}$, $\tau_m = |Pa(m)|$, and $C_m = [P_m^\top, x_{m,i}^\top]$. The theorem assumes $x_{m,i} = w_m P_m$ with linearly independent parent rows; coefficients are fixed within this batch but may change between batches. Let $l_k = |Root_k|$, $l = \max_k l_k$, and $r_k = \text{rank}(X_i(\pi_k, :))$. For $q \geq 1$, an orthonormal basis $U \in \mathbb{R}^{T \times q}$ has incoherence $\mu(U) = (T/q) \max_t \|U^\top e_t\|_2^2$, where e_t is a standard basis vector. The bound μ covers all parent row spaces and, for the comparison, the global row space. Sampling selects distinct time steps uniformly without replacement, independently of the fixed data. Following the main paper, $\Omega_m \in \{0, 1\}^{1 \times T}$ records the sampled time steps; with $D_m = \text{diag}(\Omega_m)$, $x_{m,i,\Omega_m} = x_{m,i} D_m$ and the sampled parent matrix is $P_m D_m$. All logarithms are natural.

Step 1: characterize the local causal rank. The first step identifies the dimension that determines a child's sampling requirement from its local parent relationships.

LEMMA A.2 (LOCAL CAUSAL RANK AND ROOT SPAN). *Under the theorem's assumptions, every child m in $\mathcal{G}_k$ satisfies $\text{rank}(C_m) = \tau_m$ and*

$$\tau_m \leq r_k \leq l_k \leq l, \quad \tau \leq \max_k r_k \leq r. \quad (1)$$

PROOF. The τ_m parent rows are linearly independent. The relation $\mathbf{x}_{m,i} = \mathbf{w}_m \mathbf{P}_m$ makes the child row a linear combination of these rows, so appending it adds no independent direction and $\text{rank}(\mathbf{C}_m) = \tau_m$. Because $\mathbf{C}_m^\top$ is a row submatrix of $\mathbf{X}_i(\pi_k, :)$, which is itself a row submatrix of $\mathbf{X}_i$, we obtain $\tau_m \leq r_k \leq r$.

Every metric in a DAG lies in the span of its root rows. This holds for roots directly; for each child, substitute the representations of its parents, which precede it in causal order. Induction gives $r_k \leq l_k \leq l$, and taking maxima proves the remaining inequalities. The rank equality uses linear independence of the parent data; conditional independence of variables alone does not establish it. $\square$

Step 2: preserve the local rank under sampling. Given the local rank from Lemma A.2, we next bound the samples needed to preserve it, following the concentration argument used in subspace-based matrix completion [1].

PROPOSITION A.3 (RANK PRESERVATION BY RANDOM SAMPLING). *Let $\mathbf{B} \in \mathbb{R}^{p \times T}$ have rank $q \geq 1$ and row-space incoherence at most μ . For a uniform sample of d distinct time steps, let $\mathbf{D} = \text{diag}(\Omega)$. If*

$$d \geq \min\{T, \lceil 8\mu q \log(q/\varepsilon) \rceil\}, \quad 0 < \varepsilon < 1, \quad (2)$$

then $\text{rank}(\mathbf{B}\mathbf{D}) = q$ with probability at least $1 - \varepsilon$.

PROOF. Full sampling preserves rank deterministically, so assume $d < T$. Take a compact SVD $\mathbf{B} = \mathbf{L}\mathbf{S}\mathbf{V}^\top$ and set $\mathbf{Z} = \mathbf{V}^\top \in \mathbb{R}^{q \times T}$. Since $\mathbf{Z}\mathbf{Z}^\top = \mathbf{I}_q$ and $\mathbf{L}\mathbf{S}$ has full column rank, it suffices to prove that the sampled matrix $\mathbf{S} = \mathbf{Z}\mathbf{D}$ retains rank q . Writing $\mathbf{z}_t = \mathbf{Z}(:, t)$, its Gram matrix is

$$\mathbf{Y} = \mathbf{S}\mathbf{S}^\top = \sum_{t=1}^T \Omega(t) \mathbf{z}_t \mathbf{z}_t^\top. \quad (3)$$

Each time step has inclusion probability d/T , so $\mathbb{E}[\mathbf{Y}] = (d/T)\mathbf{I}_q$ and $\lambda_{\min}(\mathbb{E}[\mathbf{Y}]) = d/T$. Moreover, incoherence bounds each summand by

$$\lambda_{\max}(\mathbf{z}_t \mathbf{z}_t^\top) = \|\mathbf{z}_t\|_2^2 \leq \mu q/T. \quad (4)$$

For comparison, let $\tilde{\mathbf{Y}} = \sum_{j=1}^d \mathbf{z}_{J_j} \mathbf{z}_{J_j}^\top$, where J_j are independent uniform draws from $\{1, \dots, T\}$ with replacement. Its expectation and summand bound are the same as those of $\mathbf{Y}$, so the matrix Chernoff inequality in Lemma A.5 applies. For sampling without replacement, the trace-exponential comparison of Gross and Nesme [2] gives $\mathbb{E} \text{tr} \exp(-\theta \mathbf{Y}) \leq \mathbb{E} \text{tr} \exp(-\theta \tilde{\mathbf{Y}})$ for every $\theta > 0$. The same Chernoff argument therefore yields

$$\Pr\{\lambda_{\min}(\mathbf{Y}) \leq (1 - \eta)d/T\} \leq q \exp\left(-\frac{\eta^2 d}{2\mu q}\right), \quad 0 < \eta < 1. \quad (5)$$

Taking $\eta = 1/2$ yields

$$\Pr\{\lambda_{\min}(\mathbf{Y}) \leq d/(2T)\} \leq q \exp\left(-\frac{d}{8\mu q}\right) \leq \varepsilon. \quad (6)$$

Outside this event, $\lambda_{\min}(\mathbf{Y}) > 0$, hence $\text{rank}(\mathbf{S}) = q$ and $\text{rank}(\mathbf{B}\mathbf{D}) = q$.

Alternatively, independent indicators $\Omega(t) \sim \text{Bernoulli}(d/T)$ give the same expectation and summand bound, so Lemma A.5 applies directly. In that formulation, d is the expected sample count; the without-replacement formulation above fixes the count at d . $\square$

Step 3: establish recovery in the causal graph. Rank preservation allows the samples to determine the causal coefficients uniquely. The following lemma proves this recovery property and examines the three cases according to whether a vector lies in the parent span, the DAG's root span, or neither. Only the parent-span case is needed for the theorem; the other two explain residual detection when the assumed representation does not hold.

LEMMA A.4 (CAUSAL RECOVERY AND RESIDUAL DETECTION). *Fix a DAG $\mathcal{G}_k$ and an exact parent matrix P_m of rank $\tau_m \geq 1$. Let $\mathcal{R}_k$ be the span of the DAG's root rows, so the row space of P_m is contained in $\mathcal{R}_k$. For a candidate row $\mathbf{x} \in \mathbb{R}^{1 \times T}$, define $\mathbf{B} = [P_m; \mathbf{x}]$, $q = \text{rank}(\mathbf{B})$, and let μ_B denote its row-space incoherence. For $0 < \varepsilon < 1$, suppose $d \geq \min\{T, \lceil 8\mu_B q \log(q/\varepsilon) \rceil\}$ time steps are sampled uniformly without replacement. With probability at least $1 - \varepsilon$, the following conclusion holds for the applicable case:*

- (1) *If $\mathbf{x} \in \mathcal{R}_k$ but $\mathbf{x}$ is outside the row space of P_m , then $\text{rank}(\mathbf{B}\mathbf{D}) = \tau_m + 1$ and the sampled least-squares residual is strictly positive.*
- (2) *If $\mathbf{x} = \mathbf{w}P_m$, then $\text{rank}(\mathbf{B}\mathbf{D}) = \tau_m$; the sampled data uniquely determine $\mathbf{w}$ and reconstruct $\mathbf{x}$.*
- (3) *If $\mathbf{x} \notin \mathcal{R}_k$, then $\text{rank}(\mathbf{B}\mathbf{D}) = \tau_m + 1$ and the sampled least-squares residual is strictly positive.*

PROOF. Write $\mathbf{A} = P_m\mathbf{D}$, $\mathbf{y} = \mathbf{x}\mathbf{D}$, and $\widehat{\mathbf{w}} = \mathbf{y}\mathbf{A}^\dagger$. Proposition A.3 gives the event $\text{rank}(\mathbf{B}\mathbf{D}) = \text{rank}(\mathbf{B}) = q$ with probability at least $1 - \varepsilon$.

Case 1: inside the DAG span but outside the parent span. Appending $\mathbf{x}$ increases rank from τ_m to $\tau_m + 1$. On the rank-preservation event, $\mathbf{y}$ cannot lie in the row space of $\mathbf{A}$, whose rank is at most τ_m . Consequently, $\|\mathbf{y} - \widehat{\mathbf{w}}\mathbf{A}\|_2^2 > 0$. Here the sampled matrix must preserve the augmented rank $q = \tau_m + 1$; its incoherence need not equal that of the parents alone.

Case 2: inside the parent span. The identity $\mathbf{x} = \mathbf{w}P_m$ implies $\mathbf{y} = \mathbf{w}\mathbf{A}$ for every sampling vector, so the sampled residual is always zero. Since $\mathbf{B}\mathbf{D} = [\mathbf{A}; \mathbf{w}\mathbf{A}]$, rank preservation also makes $\mathbf{A}$ full row rank. Its pseudoinverse is therefore

$$\mathbf{A}^\dagger = \mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1}, \quad \mathbf{A}\mathbf{A}^\dagger = \mathbf{I}_{\tau_m}. \quad (7)$$

It follows that $\widehat{\mathbf{w}} = \mathbf{w}\mathbf{A}\mathbf{A}^\dagger = \mathbf{w}$, and hence

$$\widehat{\mathbf{x}} = \mathbf{x}\mathbf{D}(\mathbf{P}_m\mathbf{D})^\dagger \mathbf{P}_m = \mathbf{w}\mathbf{P}_m = \mathbf{x}. \quad (8)$$

This is the coefficient recovery used in Equation (7) of the main paper. In this case $q = \tau_m$ and the augmented and parent row spaces coincide, so $\mu_B \leq \mu$.

Case 3: outside the DAG span. Because the parent row space is contained in $\mathcal{R}_k$, $\mathbf{x} \notin \mathcal{R}_k$ also implies that $\mathbf{x}$ is outside the parent row space. The augmented matrix therefore has rank $\tau_m + 1$, and the same argument as in Case 1 gives a positive sampled residual. For an arbitrary fixed outlier, this conclusion requires sampling to preserve the augmented rank; it is not a probability-one claim.

The residual statements use a zero threshold in exact arithmetic. A positive practical tolerance additionally requires a lower bound on the residual magnitude. Cases 1 and 3 detect failure of the specified representation; neither alone establishes that the metric is a root of the SCS. $\square$

PROOF OF THEOREM A.1. (1) *Sampling bound and reconstruction.* Take $c = 8$; a larger universal constant also suffices. For a child m in one DAG $\mathcal{G}_k$, Proposition A.3 preserves the rank of its parent matrix with probability at least $1 - \varepsilon$ when $d \geq 8\mu\tau_m \log(\tau_m/\varepsilon)$, subject to integer rounding and full-sampling capping. Case 2 of Lemma A.4 then recovers the current coefficients and the complete child vector from exact parent data. By Lemma A.2, $\tau_m \leq l_k$ and $\tau_m \leq \tau$, so $c\mu\tau \log(l_k/\delta)$ suffices for any fixed child within this DAG at failure tolerance δ .

Across K DAGs, choose $\varepsilon = \delta/K$ and use $l_k \leq l$:

$$8\mu\tau_m \log(K\tau_m/\delta) \leq c\mu\tau_m \log(Kl/\delta) \leq c\mu\tau \log(Kl/\delta). \quad (9)$$

The theorem's budget thus gives each fixed child success probability at least $1 - \delta/K \geq 1 - \delta$. Choosing $d = \min\{T, \lceil c\mu\tau \log(Kl/\delta) \rceil\}$ handles integer counts and full sampling, which is deterministic. Equation (9) also justifies using τ_m in the per-child frequency function in Equation (8) of the main paper under the randomized sampling model. A union bound covers one predetermined child from each DAG at joint probability $1 - \delta$; the simultaneous guarantee for all children is stated separately below.

(2) *Comparison with the global-rank bound.* The matrix $C_m^\top$ is a row submatrix of $X_i(\pi_k, \cdot)$, which is itself a row submatrix of X_i . Thus $\tau_m \leq r_k \leq r$ and $\tau \leq \max_k r_k \leq r$. Under the additional condition $Kl \leq r$, both factors are bounded:

$$c\mu\tau \log(Kl/\delta) \leq c\mu r \log(Kl/\delta) \leq c\mu r \log(r/\delta). \quad (10)$$

The comparison remains valid after rounding and capping at T . The causal rank can be much smaller than the global rank, but a strict improvement is not guaranteed for every batch. For $K = 1$, the bound reduces to $c\mu\tau \log(l/\delta)$ and the comparison holds under $l \leq r$. For $K = M$, every DAG is a singleton: no child metrics remain, and the online framework uses temporal SMC for every metric. This proves both statements of Theorem A.1. $\square$

LEMMA A.5 (MATRIX CHERNOFF LOWER-TAIL BOUND). *Let $Y = \sum_j Y_j$ be a sum of independent positive semidefinite $q \times q$ matrices with $\lambda_{\max}(Y_j) \leq L$ almost surely, where $L > 0$. Writing $a = \lambda_{\min}(\mathbb{E}[Y])$, for $0 < \eta < 1$,*

$$\Pr\{\lambda_{\min}(Y) \leq (1 - \eta)a\} \leq q \left(\frac{e^{-\eta}}{(1 - \eta)^{1-\eta}} \right)^{a/L} \leq qe^{-a\eta^2/(2L)}. \quad (11)$$

This is the lower-tail matrix Chernoff inequality of Tropp [3, Corollary 5.2 and Remark 5.3].

Whole-batch recovery. The theorem above concerns a child given exact parent data. To recover a complete batch, Algorithm 1 processes each DAG in causal order, fully collects roots, and then fits each child from sampled entries and reconstructed parents. The root cost is separate from the child budget. The algorithm returns FAILURE if a sampled parent matrix loses rank; the following result bounds this probability jointly over all children.

COROLLARY A.6 (CORRECTNESS OF ALGORITHM 1). *Under the stated exact linear relations, linear independence of parent vectors, and incoherence assumptions, let $N = M - \sum_k l_k \geq 1$ be the number of children and let $0 < \delta < 1$. If Algorithm 1 uses a budget $1 \leq d_m \leq T$ satisfying*

$$d_m \geq \min\{T, \lceil 8\mu\tau_m \log(N\tau_m/\delta) \rceil\} \quad (12)$$

for every child, then it returns $\hat{X}_i = X_i$ with probability at least $1 - \delta$.

PROOF. For analysis, assign every child its uniformly drawn sampling vector in advance, including children not reached if the algorithm stops. This is equivalent to drawing the vectors as they are needed in line 7, since the budgets are fixed and the draws do not depend on sampled values. For each child, define the event $\mathcal{E}_m = \{\text{rank}(P_m D_m) = \tau_m\}$ using its true parent matrix P_m . Proposition A.3, with $q = \tau_m$ and $\varepsilon = \delta/N$, gives $\Pr(\mathcal{E}_m^c) \leq \delta/N$; if $d_m = T$, the rank is preserved deterministically. Thus the event $\mathcal{E} = \bigcap_{m \text{ child}} \mathcal{E}_m$ satisfies

$$\Pr(\mathcal{E}) \geq 1 - \sum_{m \text{ child}} \Pr(\mathcal{E}_m^c) \geq 1 - \delta. \quad (13)$$

The union bound requires no independence between these events.

On $\mathcal{E}$, we prove correctness by induction along each order π_k . Every root is collected in full and is therefore exact (line 5). Suppose all metrics preceding a child m have been reconstructed correctly. Since π_k is a causal order, all its parents precede it, so $\hat{P}_m = P_m$ in line 9. The event $\mathcal{E}_m$

Algorithm 1 Offline Causal Matrix Completion

Require: Query access to entries of $X_i \in \mathbb{R}^{M \times T}$; SCS $\mathcal{G}$ with orders π_k and parent sets $Pa(m)$; child sample budgets $1 \leq d_m \leq T$.

Ensure: Reconstructed matrix $\hat{X}_i$, or FAILURE.

- 1: Initialize $\hat{X}_i = \mathbf{0}$.
- 2: **for** each DAG $\mathcal{G}_k$ **do**
- 3: **for** each metric m in order π_k **do**
- 4: **if** $Pa(m) = \emptyset$ **then**
- 5: Collect all T values and set $\hat{X}_i(m, :) = X_i(m, :)$.
- 6: **else**
- 7: Draw d_m distinct time steps uniformly without replacement; encode them in Ω_m .
- 8: Set $D_m = \text{diag}(\Omega_m)$; collect $y_m = x_{m,i} D_m$.
- 9: Set $\hat{P}_m = \hat{X}_i(Pa(m), :)$ and $A_m = \hat{P}_m D_m$.
- 10: **if** $\text{rank}(A_m) < |Pa(m)|$ **then**
- 11: **return** FAILURE.
- 12: **end if**
- 13: $\hat{w}_m = y_m A_m^\dagger$.
- 14: $\hat{X}_i(m, :) = \hat{w}_m \hat{P}_m$.
- 15: **end if**
- 16: **end for**
- 17: **end for**
- 18: **return** $\hat{X}_i$.

makes $A_m = P_m D_m$ full row rank, so the check in line 10 passes. Case 2 of Lemma A.4 then gives $\hat{w}_m = w_m$ and $\hat{X}_i(m, :) = X_i(m, :)$ in lines 13–14. Induction over all DAGs shows that the algorithm reaches line 18 and returns $\hat{X}_i = X_i$ on $\mathcal{E}$, which has probability at least $1 - \delta$. $\square$

Each completed run collects $T \sum_k l_k + \sum_m \text{child } d_m$ entries; a run that reports failure may collect fewer. The child budget therefore does not describe the cost of every row. If $N = 0$, all metrics are roots and the algorithm recovers the matrix deterministically by full sampling.

Scope of the comparison. The exact DAG relations imply $r \leq \sum_k l_k \leq Kl$. Thus $Kl \leq r$ is a restrictive sufficient condition, requiring independent root trajectories and equal root counts across DAGs; it does not follow from sparsity alone. Without it, the local and global budgets can still be compared at the same failure tolerance ε using $\tau_m \leq r$: $8\mu\tau_m \log(\tau_m/\varepsilon) \leq 8\mu r \log(r/\varepsilon)$. These are sufficient budgets for known subspaces, not necessary sample counts or claims of universal optimality.

Extension to sampled roots. If roots instead use temporal SMC, their recovery events must also be included. Let $q_m = \text{rank}(H_m)$ for an exactly known historical matrix, and assume the new root lies in its column span. With incoherence at most μ , assign roots the budget $\min\{T, \lceil 8\mu q_m \log(Mq_m/\delta) \rceil\}$ for $q_m \geq 1$, and children the corresponding budget with q_m replaced by τ_m . Proposition A.3, applied to $H_m^\top$, preserves its sampled rank. Every sampled least-squares solution then gives the same root trajectory, since $\ker(D_m H_m) = \ker(H_m)$. A union bound over at most M recovery events followed by topological induction gives whole-batch recovery with probability at least $1 - \delta$. A zero-rank historical space contains only the zero vector and needs no samples under this membership assumption. Counting only changes of the sampling vector cannot replace this argument: different parent spaces can fail on the same sample set.

Sampling conditions. The results above assume uniform random sampling, exact parent data or correctly reconstructed parents, and the stated linear relations. They do not automatically extend to the equal-interval schedule or to anomalies outside the parent span. Additional samples preserve an already full-rank representation, but do not establish subspace membership by themselves.

B Anomaly-Model Learning and Updating

This appendix derives the approximate EM updates used by the Anomaly Sampler and Model Updater (Sections 4.3 and 4.6). The SCS fixes the allowed source metrics $m' \in Pa(m) \cup \{m\}$; only their propagation strengths and decay rates are learned. We use the exponential kernel $g_{mm'}(v) = a_{mm'} \beta_{mm'} e^{-\beta_{mm'} v}$ for $v > 0$ and the conditional event rate in Equation (9).

B.1 Likelihood and Event-Source Probabilities

Let h denote the final observed time step: $h = WT$ during offline learning and $h = iT$ after online batch i . The modeled history begins at time 0, and $E_m(h+1)$ includes all detected events of metric m through time h . For the continuous-time Hawkes model evaluated at these event times, the log-likelihood is

$$\ell(\theta) = \sum_{m=1}^M \left[\sum_{s \in E_m(h+1)} \log \lambda_m(s) - \int_0^h \lambda_m(t) dt \right]. \quad (14)$$

The event terms reward rates that explain the detected anomalies, while the integrated rate also accounts for time without events. For the exponential kernel, the integrated rate is

$$\int_0^h \lambda_m(t) dt = u_m h + \sum_{m' \in Pa(m) \cup \{m\}} a_{mm'} B_{mm'}(\beta_{mm'}; h), \quad (15)$$

where the finite-horizon contribution of source metric m' is

$$B_{mm'}(\beta; h) = \sum_{v \in E_{m'}(h+1)} \left(1 - e^{-\beta(h-v)} \right). \quad (16)$$

Following the branching interpretation of Hawkes processes [4], each target event has an unobserved source: spontaneous occurrence or triggering by an earlier self or parent event. With the parameters fixed at their current values, the E-step assigns the source probabilities

$$p_{ss}^m = \frac{u_m}{\lambda_m(s)}, \quad p_{sv}^{mm'} = \frac{g_{mm'}(s-v)}{\lambda_m(s)}, \quad v \in E_{m'}(s). \quad (17)$$

The allowed sources satisfy $m' \in Pa(m) \cup \{m\}$, and $E_{m'}(s)$ contains events strictly before s . These probabilities sum to one for each target event when $\lambda_m(s) > 0$.

B.2 Offline Parameter Updates

At each offline iteration, we recompute the source probabilities and the following statistics over the training events:

$$S_{u_m} = \sum_{s \in E_m(h+1)} p_{ss}^m, \quad S_{a_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}. \quad (18)$$

$$S_{\beta_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'} (s-v). \quad (19)$$

Here S_{u_m} is the expected number of spontaneous events, $S_{a_{mm'}}$ is the expected number triggered by metric m' , and $S_{\beta_{mm'}}$ is their probability-weighted total triggering delay.

Holding these probabilities fixed, the expected complete-data log-likelihood separates into a background term $Q_{u_m} = S_{u_m} \log u_m - u_m h$ and, for each allowed source–target pair,

$$Q_{mm'} = S_{a_{mm'}} (\log a_{mm'} + \log \beta_{mm'}) - \beta_{mm'} S_{\beta_{mm'}} - a_{mm'} B_{mm'}(\beta_{mm'}; h). \quad (20)$$

Terms independent of the updated parameters are omitted. The background update is $u_m = S_{u_m}/h$. For an identifiable source–target pair, the exact finite-horizon stationary conditions are

$$a_{mm'} = \frac{S_{a_{mm'}}}{B_{mm'}(\beta_{mm'}; h)}, \quad (21)$$

$$\frac{S_{a_{mm'}}}{\beta_{mm'}} - S_{\beta_{mm'}} - a_{mm'} \sum_{v \in E_{m'}(h+1)} (h-v) e^{-\beta_{mm'}(h-v)} = 0. \quad (22)$$

Thus, exact finite-horizon updates couple the propagation strength and decay rate.

CauSample uses a boundary approximation to obtain explicit updates. Writing $n_{m'} = |E_{m'}(h+1)|$, we replace $B_{mm'}(\beta; h)$ by $n_{m'}$ and omit its derivative with respect to β . With the E-step statistics fixed, the approximate objective is $\tilde{Q}_{mm'} = S_{a_{mm'}} (\log a_{mm'} + \log \beta_{mm'}) - \beta_{mm'} S_{\beta_{mm'}} - a_{mm'} n_{m'}$. Setting its first derivatives, and that of Q_{u_m} , to zero gives

$$\frac{S_{u_m}}{u_m} - h = 0, \quad \frac{S_{a_{mm'}}}{a_{mm'}} - n_{m'} = 0, \quad \frac{S_{a_{mm'}}}{\beta_{mm'}} - S_{\beta_{mm'}} = 0. \quad (23)$$

For positive, identifiable statistics these objectives have negative second derivatives, so the solutions maximize the approximate objective. The resulting approximate M-step is

$$u_m = \frac{S_{u_m}}{h}, \quad a_{mm'} = \frac{S_{a_{mm'}}}{n_{m'}}, \quad \beta_{mm'} = \frac{S_{a_{mm'}}}{S_{\beta_{mm'}}}. \quad (24)$$

These ratios apply when the denominators are positive and the event statistics identify the corresponding parameters. For a source–target pair with no positive triggering responsibility, $S_{a_{mm'}} = S_{\beta_{mm'}} = 0$; its decay rate is then unidentifiable, and the corresponding ratio does not apply. The approximation requires both the omitted kernel tails $\sum_v e^{-\beta_{mm'}(h-v)}$ and their derivative contributions to be small relative to the retained terms. A long observation horizon alone does not ensure this condition when many source events occur near its end. Consequently, these are approximate EM updates, rather than exact M-steps for the finite-horizon objective in Equation (20). We repeat the offline E- and M-steps with $h = WT$ until the maximum parameter change falls below ϵ .

B.3 Incremental Updates after Each Batch

We initialize the cumulative statistics from the final offline E-step. After batch i , each newly sampled anomaly at time s in metric m undergoes one E-step followed by one approximate M-step. Its source probabilities are computed using the current parameters and only events strictly before s , then added to the cumulative statistics:

$$S_{u_m} \leftarrow S_{u_m} + p_{ss}^m, \quad S_{a_{mm'}} \leftarrow S_{a_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}, \quad (25)$$

$$S_{\beta_{mm'}} \leftarrow S_{\beta_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}(s-v). \quad (26)$$

For $m' \in Pa(m) \cup \{m\}$, we update the corresponding parameters using Equation (24) with $h = iT$ and $n_{m'} = |E_{m'}(iT+1)|$. Both quantities include the initial training period and all subsequent batches. They remain fixed while processing the new events from the current batch; each parameter

update precedes the E-step for the next event. The resulting parameters are used for predictions in the next batch.

Past source probabilities remain fixed, including contributions from batches that have left the sliding window. This cumulative update avoids repeated EM iterations over old target events, although each new target event still uses earlier parent and self events to compute its source probabilities. It does not claim exact maximization or monotonic improvement of the full-history likelihood. If a batch contains no sampled anomalies, no parameter update is performed; its elapsed time is included when the next event triggers an update. Within-batch predictions incorporate new events with fixed parameters, whereas parameter learning takes place after the batch ends.

References

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